

Micro I, class 23

The main event was an informal definition of an extensive form game. The first three examples of extensive form games were of this form: player 1 moves first; after observing player 1's choice, player 2 chooses. The third of these was the *Stackelberg duopoly game*, in which one firm chooses its output first; after observing that firm's choice, the other firm chooses its output. Besides its use in Industrial Organization, this leader-follower structure arises often in economics. The "vertical integration" problem in ASMT VII is an example. The Principal-Agent model (which you will take up in Micro II) is another.

As I mentioned, beginners in game theory often struggle with writing down strategies in extensive form game. Consider this general "Stackelberg" game, which, as mentioned, includes all three of today's examples. First player 1 chooses an action $a_1 \in A_1$; after observing a_1 , player 2 then chooses an action $a_2 \in A_2$. After that the game ends. The set of terminal histories is $T = A_1 \times A_2$. Preferences on T are represented by $U_i : T \rightarrow \mathbb{R}$. A strategy for player 1 is just a point $a_1 \in A_1$, and its strategy set is just $S_1 = A_1$. A strategy for player 2 is a *function* $s_2 : A_1 \rightarrow A_2$. Its strategy set is the set of *all* such functions. Formally

$$S_2 = \{s_2 \subseteq A_1 \times A_2 \mid (a_1, a_2) \in s_2 \text{ and } (a_1, a'_2) \in s_2 \Rightarrow a'_2 = a_2\}.$$

This formal set-theoretic definition is useful to keep in mind in coming up with a clear notation for strategies.

I also introduced the notion of a *subgame perfect Nash equilibrium* (SPNE): it is a NE that induces a NE on every subgame. I will present the formal definition of a subgame on Thursday. But in the context of the three examples I presented in class, a subgame begins at each move. In those examples, at each move, a player knows all previous actions. These are called *perfect-information* games. My definition of a subgame works for all such games (including problems #1 and #2 on ASMT X).

I will next introduce the formal definition of an extensive form game and of a SPNE. After some more examples, time permitting, I will begin a discussion of repeated games, our final topic, using the prisoners' dilemma as a leading example.